Housing matches with hidden options

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Our starting point: brokers can affect which apartments people get to consider. We want to understand what that does to matching.

Work in progress, for discussion

The broker idea, simplified

Suppose a renter and a landlord would both prefer to match, but their agents never put them in contact.

We start with renters and listings as the two sides. Each has a fixed agent who controls access to some of its possible matches.

An *audit* opens the feasible matches involving that agent's clients. It has a cost.

This leaves out the reason an agent hides an option. We have not modeled commissions, friendships, or collusion yet.

What we are assuming

E contains all feasible renter–listing pairs. Initially, only $E_0 \subseteq E$ is available. Preferences are strict and fixed.

Auditing a set of agents A gives

$$E(A) = E_0 \cup \{r\ell \in E : f(r) \in A \text{ or } f(\ell) \in A\},$$

where $f(i)$ is client i's agent. The platform then runs

$$D(A) = \mathrm{DA}_R(E(A)).$$

We want the cheapest A for which no pair in *all of* E would rather match with each other than keep its assignment.

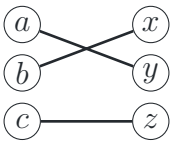
DA = renter-proposing deferred acceptance. We assume the planner knows E and the rankings but pays to make pairs available. We still need a housing setting that justifies these assumptions.

A small example

Three renters, three listings. Start with $E_0 = \{ay, bx, cz\}$. Each client has its own agent, and every pair is feasible.

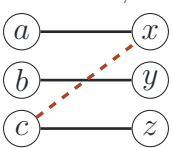
$$a : x \succ y \succ z \quad b : y \succ x \succ z \quad c : x \succ z \succ y$$
$$x : b \succ c \succ a \quad y : a \succ b \succ c \quad z : c \succ a \succ b$$

Start here



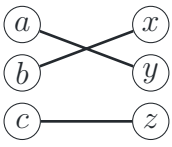
Stable

Audit a, b



cx blocks

Also audit c



Stable again

In the middle, c wants x instead of z, and x wants c instead of a.

The solid lines show DA's assignments. The dashed pair is still hidden. Preferences have not changed.

Having a stable matching available isn't enough

In the middle graph, the original stable matching $\{ay, bx, cz\}$ is still available. DA just doesn't choose it.

So there are two different costs:

$$C_{\text{support}}: \text{ make some full-market stable matching available,}$$
$$C_{\text{DA}}: \text{ make DA's chosen matching full-market stable.}$$

Starting from N copies of that middle graph, plus one hidden isolated pair,

$$C_{\text{support}} = 1, \qquad C_{\text{DA}} = N + 1.$$

Unit audit costs, one client per agent. The proof is in the appendix. It is a simple construction. We see it as a reason to study the distinction, and still need to establish what is new.

Where this sits in the literature

- Liu et al. (2024) already audit racial steering in LLM housing recommendations.
- Matching papers study costly interviews, limited information, and changes to the available pairs.
- Trabelsi et al. (2025) is especially close: pay to add pairs while a platform keeps using its own maximum-matching rule.

Our question is narrower: which agents should we audit so that DA leaves no blocking pair hidden?

We need to check whether this exact problem, or an equivalent one, has already been studied. References and links are in the appendix.

How we could test the housing part

Use one fixed catalogue of rental listings. Give an AI recommender household profiles and ask it to return listing IDs. These recommendations define the initially visible pairs.

Then compare targeted audits with random audits at the same budget, and with full disclosure.

Historical listings could supply rents and apartment features. ACS could help us generate household profiles. Preferences, agent groups, and audit costs would still need assumptions.

This study has not been run. Candidate data: Two Sigma rental listings and ACS PUMS. Keep catalogue and list length fixed, randomize listing order, and repeat prompts. The existing experiments use synthetic markets.

Stability may leave affordable options hidden

For each modeled renter r, let F_r be their affordable feasible listings. Let G^+ be the lower-income renters with at least one such option. Measure

$$\mathrm{Access}(A) = \frac{1}{|G^+|} \sum_{r \in G^+} \frac{|\{\ell \in F_r : r\ell \in E(A)\}|}{|F_r|}.$$

Let $C(\tau)$ be the cheapest audit plan that makes DA full-market stable and gives $\mathrm{Access}(A) \geq \tau$.

$$\underbrace{C(\tau) - C(0)}_{\text{extra cost of the access target}}$$

Does this cost become large when affordable housing is scarce? Does the recommender change it?

Assume $|G^+| > 0$ and $\tau \in [0, 1]$. Report renters with no affordable option separately. This measures available options, not who ultimately gets a home.

What we'd like your advice on

The biggest issue for us is the institution. Is there a housing setting where a fixed matching rule and costly access make sense?

Would you push this toward a theory result, or toward measuring the cost of affordable access with an actual recommender?

And which paper should we read before going further?

Why the costs are 1 and $N + 1$

Start each copy with

$$H = \{ax, ay, az, bx, by, bz, cz\}.$$

1. $\{ay, bx, cz\} \subseteq H$ is stable in the full market. But DA returns $\{ax, by, cz\}$, and cx blocks.
2. Without auditing c or x, the only new pair we can add is cy. It changes nothing: c prefers z, and z keeps c.
3. Auditing either c or x exposes cx. The rejection chain returns DA to $\{ay, bx, cz\}$.

So each copy costs zero for support and one for DA. A separate hidden isolated pair costs one under either objective.

A case where the distinction disappears

Suppose every listing uses the same strict priority over renters. Then there is a unique stable matching M^*: let renters choose in priority order.

$$M^* \subseteq H \subseteq E \quad \Longrightarrow \quad \mathrm{DA}_R(H) = M^*.$$

Each renter's choice from the full market is still available at their turn. So $C_{\text{support}} = C_{\text{DA}}$.

We only need to cover the hidden pairs in M^*. If agents represent one side each, this is weighted bipartite vertex cover and we can solve it by minimum cut.

This follows from standard matching and covering results. It is a useful check on which assumptions make our problem difficult.

References

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